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International Mathematics Assessments for Schools (2022-2023)

Middle Primary Division Round 2

Time: 120 minutes

Printed Name _____ *Code* _____ *Score* _____

Instructions:

- Do not open the contest booklet until you are told to do so.
- Be sure that your name and code are written on the space provided above.
- Round 2 of IMAS is composed of three parts; the total score is 100 marks.
- Questions 1 to 5 are given as a multiple-choice test. Each question has five possible options marked as A, B, C, D and E. Only one of these options is correct. After making your choice, fill in the appropriate letter in the space provided. Each correct answer is worth 4 marks. There is no penalty for an incorrect answer.
- Questions 6 to 13 are a short answer test. Only Arabic numerals are accepted; using other written text will not be honored or credited. Some questions have more than one answer, as such all answers are required to be written down in the space provided to obtain full marks. Each correct answer is worth 5 marks. There is no penalty for incorrect answers.
- Questions 14 and 15 require a detailed solution or process in which 20 marks are to be awarded to a completely written solution. Partial marks may be given to an incomplete presentation. There is no penalty for an incorrect answer.
- Use of electronic computing devices is not allowed.
- Only pencil, blue or black ball-pens may be used to write your solution or answer.
- Diagrams are not drawn to scale. They are intended as aids only.
- After the contest the invigilator will collect the contest paper.

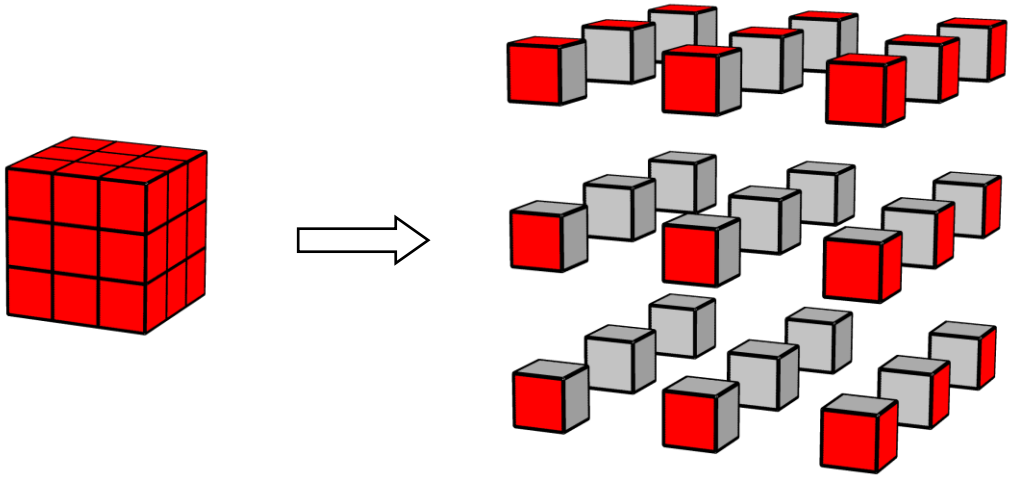
The following area is to be filled in by the judges;
the contestants are not supposed to mark anything here.

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Total Score	Signature
Score																	
Score																	

Middle Primary Division Round 2

Questions 1 to 5, 4 marks each

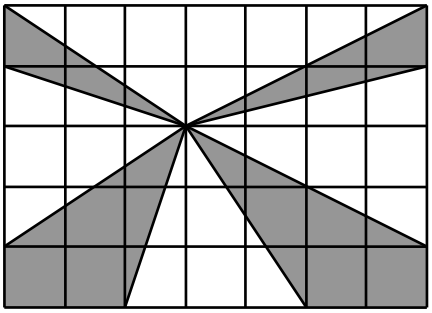
1. To paint all the six faces of a $3\times3\times3$ cube, 6 g of red paint must be used. After the paint dries up, the cube is then cut into 27 identical unit cubes, as shown in the diagram below. Determine the amount of red paint, in g, that is needed to paint all the remaining unpainted faces of these unit cubes?



（A）12　　（B）18　　（C）24　　（D）30　　（E）36

Answer：_____

2. As shown in the diagram below, there is a 5×7 grid paper where each small square has an area of 1 cm^2. What is the total area, in cm^2, of the shaded regions?



(A) 12　　(B) 13　　(C) 14　　(D) 21　　(E) 22

Answer：_____

3. In 1945, a famous mathematician said: "*At a certain point in my life, the square of my age coincided with the year in which I was that age.*" At what age did the mathematician made this comment?

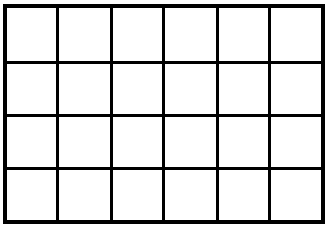
（A）44　　（B）45　　（C）46　　（D）52　　（E）53

Answer：_____

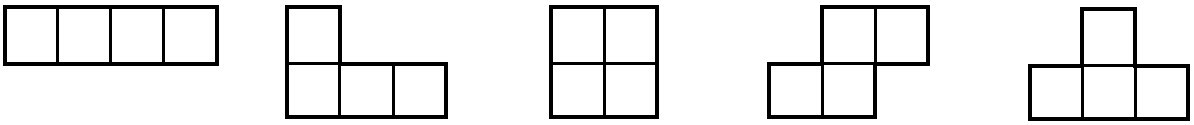
4. How many different four-digit numbers (where its first digit cannot be 0) are there that can be formed by using two “2”s, one “0” and one “3” as its digits?
（A）5　　（B）9　　（C）12　　（D）18　　（E）24

Answer：_____

5. A 4×6 rectangle is constructed using six pieces, five of which are shown as the options. The sixth piece is a duplicate of one of them. Which piece appears twice?



(A) (B) (C) (D) (E)



Answer：_____

Questions 6 to 13, 5 marks each

6. The number 5009 has the following property: no matter how one re-arranges its digits, one cannot obtain a smaller 4-digit number which does not start with a 0. What is the next 4-digit number that has this property?

Answer：_____

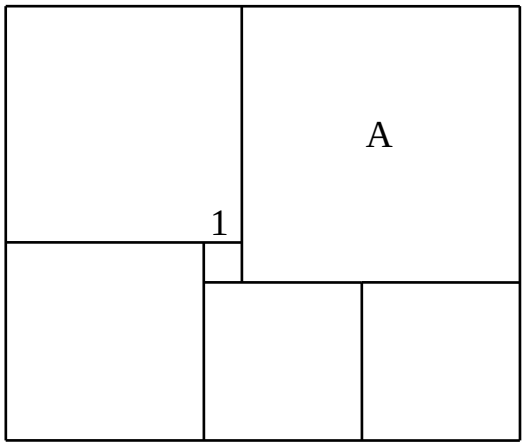
7. Alice and Bruce have the same number of tea bags, each of which may be used to make either 2 or 3 cups of tea. Alice uses all of her tea bags to make 41 cups of tea, while Bruce uses all of his tea bags to make 58 cups of tea. How many tea bags does each have initially?

Answer：_____ tea bags

8. A park has only linden and maple trees. At the start of the year, three fifths of the trees are maple trees. Some more linden trees are planted in the spring, and then only one fifth of all the trees are maple trees. In the fall, some more maple trees are planted, and once again, three fifths of all the trees are maple trees. Find the ratio of the number of trees now to their initial number.

Answer：_____ : _____

9. A rectangle is dissected into six squares as shown in the diagram below. If the side length of the smallest square is 1 cm, what is the side length, in cm, of square A?



Answer : _____ cm

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10. In the following addition:

$$\begin{array}{cccccc} & & & N & O \\ + & W & I & L & D \\ \hline & T & E & E & N \end{array}$$

where different letters represent different digits and the same letter represents the same digit. If the letters N, W and T cannot be 0, then what is the minimum possible value of $TEEN$?

Answer : _____

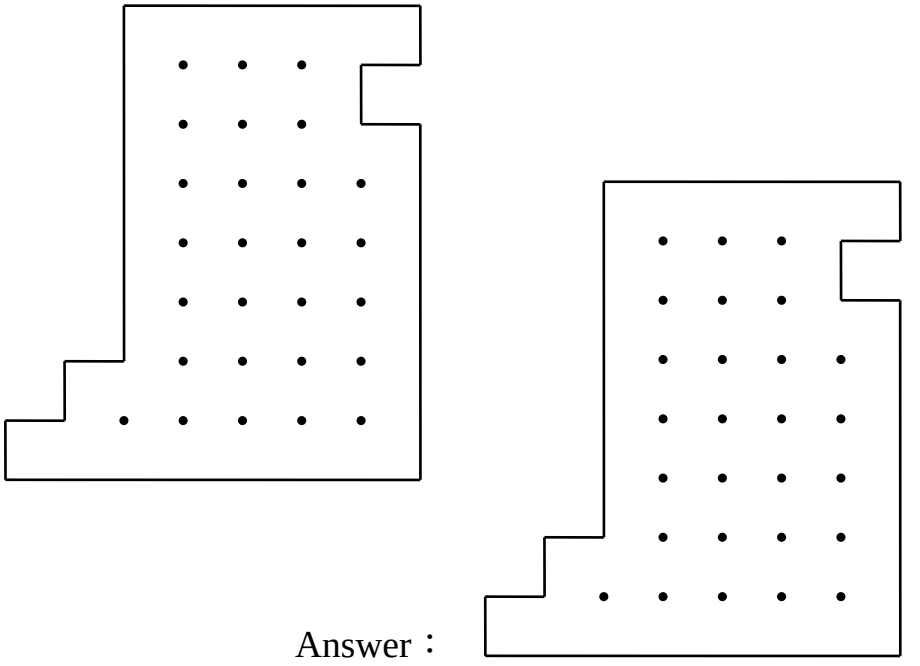
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11. How many distinct numbers can be obtained as the sum of four distinct numbers of 17, 21, 25, 29, 33, 37, 41 and 45?

Answer : _____

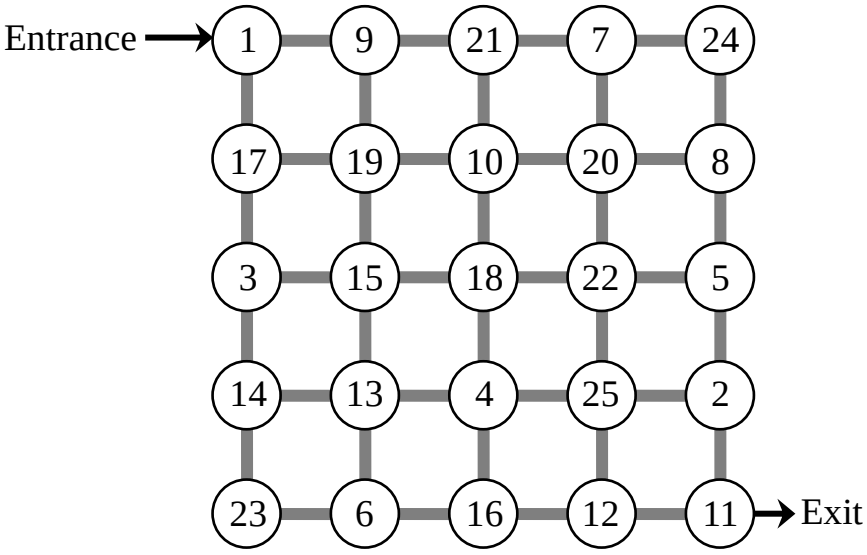
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12. Dissect the following figure into two figures of the same size and the same shape after reflection and rotation.



Answer：

13. Andy has to pass through an underground maze. The enchantment on the maze requires Andy to pick up the piece of stone in the first room when he enters the maze. Then in each subsequent room he must exchange it with the next stone if that one is heavier.



The number in each room represents the weight of the stone in it, in kg. Andy can only walk to the right or down on the map. He also wants to carry as little weight as possible when he leaves the maze.
What is the least weight, in kg, he can leave with if he only walks right or down?

Answer： _____ kg

Questions 14 to 15, 20 marks each
(Detailed solutions are needed for these two problems)

14. A Mathematics test consists of 3 problems, where each problem is being graded independently with integer points between 0 and 5. How many ways can the total points for this test be exactly 11?

Answer : _____ ways

15. In how many ways can we obtain a total score of 44 from the table shown, by following the conditions below?
    - Start with any unit square of your choice and collect further different three numbers from the other squares which touch at least one selected square by a full side only.
    - All the numbers chosen must be different and the number 12 must always be one of them.

		9				
	13	10	9			
	13	9	13	10	10	
13	9	13	12	10	13	9
	10	10	13	9	10	
		13	9	13		
			10			

The following figures below are two examples. Note that your answer must include both examples.

		9				
	13	**10**	9			
	13	9	**13**	10	10	
13	9	13	**12**	10	13	9
	10	10	13	9	10	
		13	9	13		
			10			

		9				
	13	10	9			
	13	**9**	**13**	**10**	10	
13	9	13	**12**	10	13	9
	10	10	13	9	10	
		13	9	13		
			10			

Answer： There are _____ ways, and the ways are as following:

